Quality Metrics Analysis & Process Improvement

Analytical Work Sample

Data Analysis | Data Quality | Business Insights | Process Improvement | Visualization | Recommendations

Prepared as an analytical work sample demonstrating data analysis, reporting, visualization, and process-improvement capabilities.

SEPTEMBER, 2026

Lusine Artenyan

Enhancing Quality Metrics Analysis

Executive Summary & Data Quality Assessment

Objective

This analysis evaluates 12 months of quality performance data to identify data-quality issues, performance trends, relationships among key metrics, and opportunities for improvement. The review focuses on **Error Rate, Completion Time, Customer Satisfaction, and Compliance** to support quality improvement, operational monitoring, and management decision-making.

Data Validation and Preparation

The dataset contained 12 monthly observations and was reviewed for missing values, duplicates, invalid values, and values outside reasonable business ranges. Three data-quality exceptions were identified:

Month	Metric	Source Value	Issue
March	Error Rate	Missing	Missing value
June	Customer Satisfaction	200%	Invalid percentage
August	Completion Time	-1 hr	Invalid negative duration

No duplicate monthly records were identified. All remaining percentage values were within the expected 0%–100% range, and remaining completion-time values were nonnegative.
The original data was preserved for audit traceability. For **analytical purposes only**, the three exceptions were estimated using the average of the immediately preceding and following months:

- March Error Rate: **7.98%**
- June Customer Satisfaction: **88.00%**
- August Completion Time: **36.53 hours**

In a production environment, these exceptions should be verified against the source system or responsible data owner before permanent correction. Following cleaning, the dataset was evaluated using the **1.5×IQR method**, and no additional statistical outliers were identified.

KPI Summary

Metric	Average	Median	Minimum	Maximum
Error Rate	**5.68%**	6.40%	1.19%	9.73%
Completion Time	**33.20 hrs**	32.05 hrs	27.35 hrs	43.98 hrs
Customer Satisfaction	**86.79%**	87.89%	71.95%	98.97%
Compliance	**91.05%**	90.64%	85.52%	98.64%

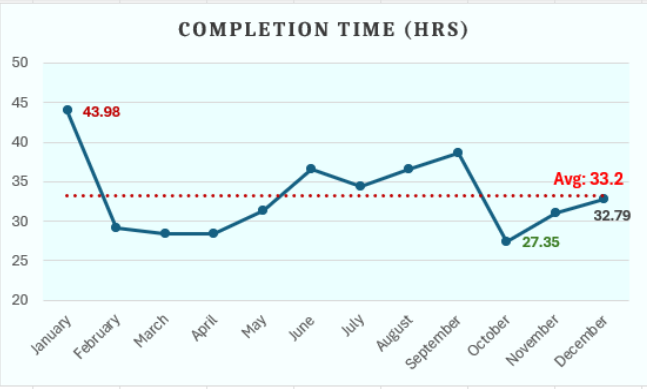
Overall Assessment

Performance varied meaningfully across all four measures. Several months demonstrated strong results, but the month-to-month fluctuations indicate that improvements were not consistently sustained and warrant further review of underlying operational factors.

Trend & Relationship Analysis

Key Performance Trends





Customer Satisfaction. Satisfaction averaged **86.79%** and declined from 88.23% in July to a low of **71.95% in September**, followed by a sharp rebound to 98.47% in October and a high of **98.97% in November**. The improvement warrants review of service changes, survey participation, respondent mix, or other factors that may have contributed to the change.

Completion Time. Completion time averaged **33.20 hours**, ranging from **43.98 hours in January** to **27.35 hours in October**. Turnaround time improved early in the year, increased from May through September, and then improved sharply from 38.68 hours in September to 27.35 hours in October. Operational changes associated with the October improvement should be reviewed for potential replication.

Additional Findings

Metric	Key Observation
Error Rate	Varied from **1.19% in November** to **9.73% in December**. Elevated results in February and August also indicate inconsistent error performance.
Compliance	Varied from **85.52% in July** to **98.64% in August**, then declined to 88.88% in September, indicating that strong performance was not consistently sustained.

Correlation Analysis

Relationship	Correlation (r)	Interpretation
Completion Time ↔ Customer Satisfaction	**-0.44**	Moderate negative relationship
Error Rate ↔ Compliance	**0.49**	Moderate positive relationship; further review warranted

Longer completion times tended to be associated with lower customer satisfaction. This relationship may indicate an opportunity to investigate whether turnaround time affects stakeholder experience.

The positive relationship between error rate and compliance is counterintuitive and should not be interpreted without additional data. The measures may reflect different aspects of performance or may be influenced by workload, timing, process changes, or other factors.

Interpretation note: The analysis includes only 12 monthly observations. Correlations should therefore be treated as **exploratory signals, not evidence of causation**.

Benchmarking

External benchmark values were not included in the provided dataset, and the metric definitions are not detailed enough to support a valid comparison with outside standards. Therefore, the **annual averages and best-performing months were used as internal reference points**. Any future comparison with DDS or other regional-center benchmarks should first confirm that metric definitions, populations, and calculation methods are equivalent.

Recommendations & Monitoring Plan

Recommended Actions

Recommendation	Action
1. Establish structured error root-cause review	Categorize errors by type, process, workload, and recurrence. Document root cause, corrective action, owner, due date, and follow-up results. Use targeted training only where knowledge or procedural gaps are confirmed.
2. Monitor turnaround time and bottlenecks	Establish turnaround targets and monitor aging, overdue work, and process stages. Review conditions associated with stronger-performing months, particularly October.
3. Strengthen satisfaction monitoring	Track satisfaction together with response count, response rate, service/process, and complaint themes. Investigate large shifts to distinguish service improvement from changes in survey participation.
4. Implement proactive compliance and corrective-action tracking	Track findings, owners, due dates, status, completion dates, and verification. Regularly review overdue and high-priority corrective actions.
5. Measure change-management effectiveness	Establish baseline measures before interventions and reassess results at appropriate follow-up intervals to determine whether improvements are sustained.

Proposed New Quality Metrics

To strengthen ongoing quality monitoring, NLACRC could consider tracking:

Metric	Purpose
First-Pass Accuracy %	Measures work completed correctly without rework
Repeat Error Rate	Identifies recurring issues after corrective action
Corrective-Action Closure Rate	Tracks completion of corrective actions
Average Days to Close Audit Findings	Measures timeliness of audit resolution
% Corrective Actions Completed on Time	Measures adherence to corrective-action deadlines
Turnaround-Time Target Attainment	Tracks the percentage of work completed within target
Backlog Aging	Identifies unresolved work by age
Survey Response Rate	Adds context to satisfaction results
Training Effectiveness	Compares performance before and after training
Post-Change Sustainability	Measures whether improvements persist over time

Monitoring Approach

Quality monitoring should connect **performance trends, audit findings, corrective actions, stakeholder feedback, and change-management results**. High-risk or overdue items should be escalated, while successful interventions should be documented and standardized where appropriate.

Measure → Identify → Investigate → Correct → Verify → Sustain

Conclusion

The analysis identified meaningful variability in quality, efficiency, customer satisfaction, and compliance. A more structured approach to root-cause analysis, corrective-action tracking, survey monitoring, and post-change evaluation would help NLACRC move from retrospective reporting toward proactive quality improvement. Additional case-level, workload, survey, and operational data would further strengthen future analysis and support more targeted improvements in service delivery.

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Enhancing Quality Metrics Analysis

Executive Summary & Improvement Recommendations

12-month review of quality performance, data quality, trends, relationships, and corrective-action monitoring.

Focus Metrics

- Quality — **Error Rate**
- Efficiency — **Completion Time**
- Experience — **Satisfaction**
- Control — **Compliance**



Prepared by: Lusine Artenyan
September 2026

Objective & Analysis Scope

What the assessment is designed to evaluate and support.

Objective

Evaluate 12 months of quality performance data to identify data-quality issues, performance trends, relationships among key metrics, and opportunities for improvement.

KPI Selection & Definitions

Metric	What It Measures	Direction
Error Rate	Frequency of errors in reported performance	Lower is better
Completion Time	Time required to complete work/turnaround	Lower is better
Customer Satisfaction	Reported stakeholder experience or survey result	Higher is better
Compliance	Adherence to required standards or processes	Higher is better

Analytical Approach

- Validate and clean source data before analysis
- Summarize KPIs using average, median, minimum, and maximum
- Use trends and correlations as exploratory signals
- Translate findings into corrective actions and monitoring metrics

Output: A concise management view that connects performance trends to follow-up actions and sustained quality improvement.

Data Quality Assessment

Original values were preserved; documented analytical estimates were used only where needed.

Three Source-Data Exceptions

Month	Metric	Source Value	Issue	Analysis Value
March	Error Rate	Missing	Missing value	7.98%
June	Customer Satisfaction	200%	Invalid percentage	88.00%
August	Completion Time	-1 hr	Invalid negative duration	36.53 hrs

Treatment: Each exception was estimated using adjacent-month interpolation for analytical purposes only.

Traceability: Original source data should remain available so any permanent correction can be verified with the source system or data owner.

Validation Conclusion

- No duplicate monthly records identified
- Remaining percentage values were within the expected 0%–100% range
- Remaining completion-time values were nonnegative
- No additional statistical outliers were identified using the 1.5×IQR method

Data preparation supports credible analysis while preserving auditability.

KPI Snapshot

Performance varied meaningfully across all four measures.

Metric	Average	Median	Minimum	Maximum	Readout
Error Rate	5.68%	6.40%	1.19%	9.73%	Inconsistent error performance
Completion Time	33.20 hrs	32.05 hrs	27.35 hrs	43.98 hrs	Turnaround variability
Customer Satisfaction	86.79%	87.89%	71.95%	98.97%	Experience volatility
Compliance	91.05%	90.64%	85.52%	98.64%	Compliance not consistently sustained

Lowest Error Rate
1.19% Nov

Best Completion Time
27.35 hrs Oct

Highest Satisfaction
98.97% Nov

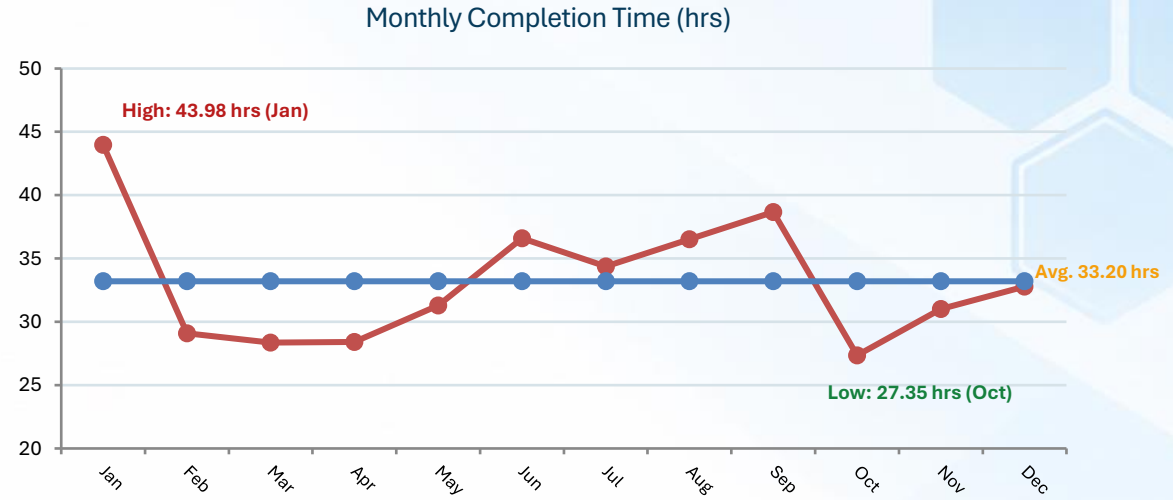
Highest Compliance
98.64% Aug

Overall assessment: Several months demonstrated strong results, but the month-to-month fluctuations indicate that improvements were not consistently sustained and warrant review of underlying operational factors.

Key Performance Trends

Customer experience and turnaround time showed notable month-to-month movement.





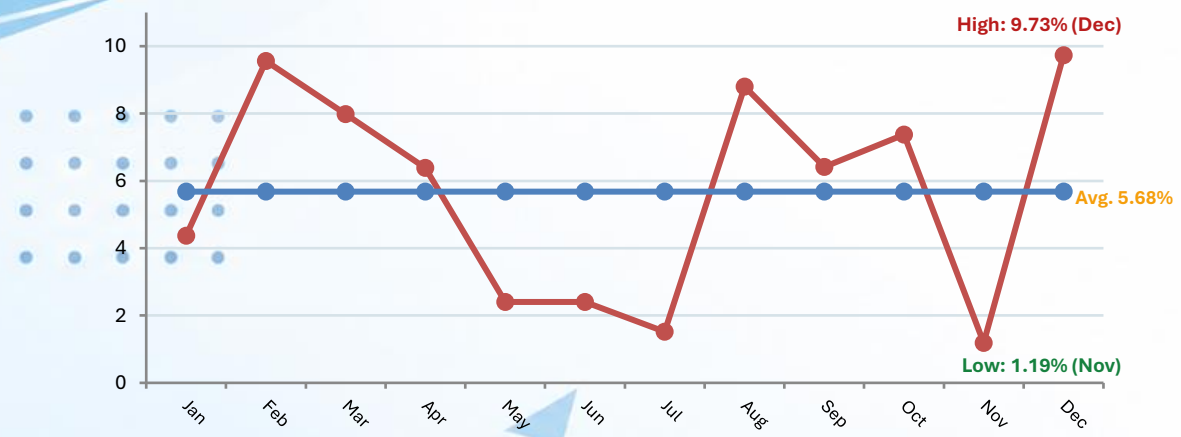
Customer satisfaction: Declined to 71.95% in September, then rebounded to 98.47% in October and 98.97% in November. Review survey participation, respondent mix, and service changes.

Completion time: Improved from 38.68 hours in September to 27.35 hours in October. Review workload, process changes, case complexity, and staffing conditions for practices that can be replicated.

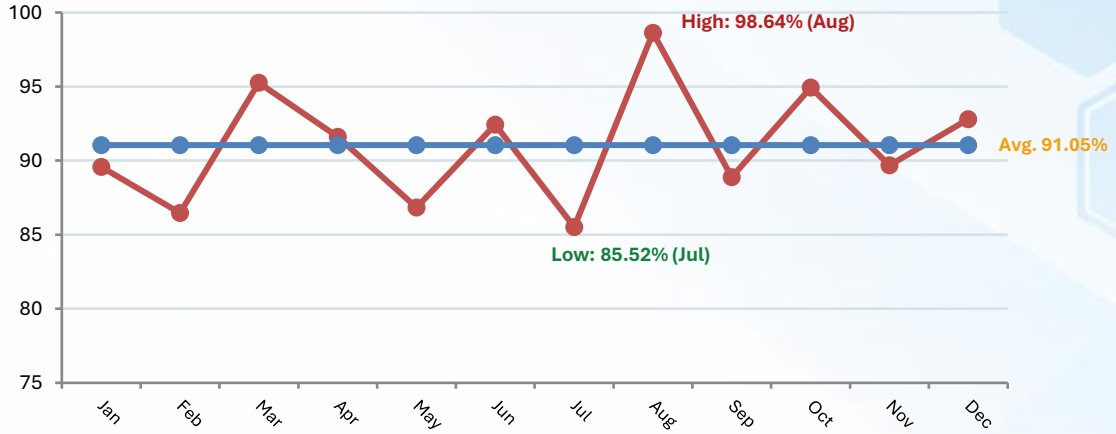
Additional Trends & Exploratory Relationships

Error-rate and compliance movement should be reviewed alongside operational context.

Monthly Error Rate (%)



Monthly Compliance (%)



Error rate: Varied from 1.19% in November to 9.73% in December. Elevated results in February and August also indicate inconsistent performance.

Compliance: Varied from 85.52% in July to 98.64% in August, then declined to 88.88% in September, indicating strong performance was not consistently sustained.

Relationship	r	Interpretation
Completion Time vs Satisfaction	-0.44	Moderate negative relationship
Error Rate vs Compliance	0.49	Counterintuitive; further review warranted

Interpretation note: Correlations use only 12 monthly observations and should be treated as exploratory signals, not evidence of causation.

Benchmarking & Internal Reference Points

External comparisons require equivalent metric definitions, populations, and calculation methods.

Internal Reference Points

Metric	Annual Average	Best Month	Best Result	Internal Benchmark Use
Error Rate	5.68%	November	1.19%	Investigate low-error practices
Completion Time	33.20 hrs	October	27.35 hrs	Replicate efficient turnaround practices
Customer Satisfaction	86.79%	November	98.97%	Sustain survey/service drivers
Compliance	91.05%	August	98.64%	Standardize high-compliance practices

Benchmarking approach: Because external benchmark values were not included in the dataset, annual averages and best-performing months were used as internal reference points.

Caution: Do not compare against DDS or regional-center benchmarks until metric definitions, populations, and calculation methods are confirmed to be equivalent.

Best-performing months should be reviewed for repeatable practices, not treated as guaranteed targets without operational context.

Recommendations & Monitoring Plan

Move from retrospective reporting to proactive quality improvement.

Recommended Actions

Recommendation	Action
1. Structured error root-cause review	Categorize errors by type, process, workload, and recurrence; document root cause and corrective action.
2. Monitor bottlenecks	Set turnaround targets and monitor aging, overdue work, and process stages.
3. Strengthen satisfaction monitoring	Track survey response counts/rates and analyze satisfaction by service or process.
4. Corrective-action tracking	Track findings, owners, due dates, completion dates, and verification.
5. Change-management effectiveness	Establish baselines and reassess results to determine whether gains are sustained.

Proposed New Metrics

- First-pass accuracy %
- Repeat error rate
- Corrective-action closure rate
- Average days to close audit findings
- % corrective actions completed on time
- Turnaround-time target attainment
- Backlog aging
- Survey response rate
- Training effectiveness
- Post-change sustainability



Monitoring approach: Connect performance trends, audit findings, corrective actions, stakeholder feedback, and change-management results. Escalate high-risk or overdue items and standardize successful interventions.

Corrective Action & Follow-Up Tracker

Translate findings into accountable ownership, targets, measures, and verification.

Finding	Priority	Proposed Action	Owner	Target	Status	Follow-up
Error-rate variability	High	Perform error-category root-cause review and identify repeat errors	Process Owner / QA Lead	30 days	Not Started	30-day review
Satisfaction decline	High	Review survey responses, response rate, and operational factors	Customer Service / Operations	30 days	Not Started	30-day review
Completion-time variability	Medium	Analyze workflow bottlenecks, backlog aging, and case complexity	Operations Manager	45 days	Not Started	30/60-day review
Compliance fluctuations	High	Review exceptions and practices that drove August performance	Compliance / QA Lead	30 days	Not Started	30/60-day review
Corrective-action follow-through	High	Establish formal closure tracking with owners and verification	QA / Management	30 days	Not Started	Monthly review

Success measures: reduced error-rate variability, improved turnaround target attainment, sustained compliance, higher satisfaction stability, and timely corrective-action closure.

Action-tracker logic: Finding → Priority → Action → Owner → Target Completion → Status → Follow-up.

This creates a direct audit trail from analysis to management follow-up and post-change verification.

Thank You

For Your Time and Attention

Key Takeaway

Meaningful variability was identified across quality, efficiency, satisfaction, and compliance. Structured root-cause analysis, corrective-action tracking, and ongoing monitoring can help convert short-term gains into sustained performance.



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September 6, 2026